Battery-Aware Multi-Sensor Exploration in Unseen Waters

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CSCE 775: Deep Reinforcement Learning

***Abstract*—Autonomous Surface Vehicles (ASVs) deployed for long-duration monitoring face a critical trade-off between maximizing information gain and conserving limited battery life. High-fidelity sensors like Radar and Lidar provide rich data but consume significant power, while environmental factors like currents and waves impose varying energy costs on propulsion. This proposal studies an environment-aware reinforcement learning policy that jointly optimizes motion control and sensor duty-cycling. By conditioning the policy on inferred water-body types—such as lake, river, or coast—the agent learns distinct strategies to maximize coverage efficiency. We compare the proposed method against standard frontier exploration, rule-based coverage, and energy-unaware RL baselines, evaluating performance in terms of area mapped per Joule and mission success rate under diverse hydrodynamic conditions.**

I. Introduction

Autonomous Surface Vehicles (ASVs) are increasingly used for marine monitoring, but mission duration is constrained by finite battery and high sensing/propulsion cost. Exploration methods that maximize coverage without energy modeling can terminate early or fail to return safely, especially in dynamic water conditions.

This proposal studies whether an ASV can jointly optimize motion and sensing to maximize information gain per Joule. We propose an RL policy that controls both vehicle motion and sensor duty-cycling, explicitly conditioned on environment type (‘lake‘, ‘river‘, ‘coast‘). The hypothesis is that environment-aware joint control outperforms fixed-sensor frontier methods and energy-unaware RL baselines.

The design is grounded in exploration and information-gathering literature [1], [2], and in recent energy-aware marine planning advances [3]–[5]. Unlike prior approaches that separate path planning and sensor scheduling, our method learns both in one policy loop (Fig. **??**) and evaluates the resulting capability gap against related work (Fig. **??**).

II. Approach

We formulate exploration as a partially observable Markov decision process (POMDP). At step t, the policy receives state

$$s_t = \{\text{local map belief, sensor features, } E_t,\, w_{type}\},$$

where E_t is battery state and w_{type} is environment type.

The action is hybrid:

$$a_t = \{v_t,\, \omega_t,\, u_t^{radar},\, u_t^{lidar},\, u_t^{cam}\},$$

where (v_t, ω_t) are motion commands and $u_t^{(\cdot)} \in \{0, 1\}$ controls sensor duty-cycling.

Reward combines information gain, energy use, and safety:

$$r_t = \lambda_1 \Delta\text{InfoGain} - \lambda_2 \text{EnergyCost}(a_t, w_{type}) - \lambda_3 \text{CollisionPenalty}. \quad (1)$$

This objective encourages policies that preserve coverage quality while avoiding unnecessary high-power sensing and costly counter-current maneuvers. We train with a continuous-control actor-critic method (SAC/PPO variant) and compare against frontier, rule-based, and energy-unaware RL baselines.

III. Evaluation

Evaluation is conducted in a marine simulator with hydrodynamics and sensor-level power accounting. We define three scenario families: ‘lake‘ (low current), ‘river‘ (strong directional flow), and ‘coast‘ (wave/wind disturbance), each with held-out maps for generalization testing.

We compare five settings: (1) frontier exploration with sensors always on, (2) rule-based coverage with fixed energy heuristic, (3) energy-unaware RL, (4) energy-aware RL without environment conditioning, and (5) the proposed environment-aware policy with duty-cycling.

Primary metrics are coverage efficiency (m^2/J), mission success rate under battery constraints, and total information gain. Secondary metrics include collision rate, active-sensor duty ratio, and return-to-base reliability.

Stress tests include low initial battery (30%), abrupt environment transition (lake-to-river), and single-sensor dropouts. We report confidence intervals across random seeds.

Published quantitative evidence motivating this setup is summarized in Fig. **??**: wake-informed marine planning shows measurable energy benefits over current-informed baselines, and flow-aware deep RL reports substantial conservation gains in vortical conditions [4], [5].

IV. Conclusion

This project addresses the fundamental challenge of extending the operational endurance of autonomous marine vehicles. We propose a battery-aware reinforcement learning policy that integrates sensor management with motion planning, explicitly adapting to environmental dynamics like currents and waves. By treating energy conservation as a first-class objective alongside exploration, we aim to demonstrate that intelligent, context-aware behaviors can significantly increase the effective range and utility of ASVs.

We benchmark this approach against standard frontier and rule-based methods, as well as non-adaptive RL baselines.

The expected outcome is a policy that not only covers more area per Joule but also exhibits distinct, interpretable strategies for different water bodies—such as exploiting currents for transport or duty-cycling sensors in safe regions. If full RL training proves unstable, a fallback to a hierarchical approach with separate planning and power management modules will still provide valuable insights into the energy dynamics of marine autonomy.

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